Hrishikesh Thakur

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Software Engineer with 2+ years architecting backend systems, AI-driven applications, and data pipelines handling 30M+ records. Experienced in designing microservices and delivering production-ready solutions in cross-functional environments

Education

University of Southern California — August 2023 – May 2025
Master of Science in Computer Science – GPA : 3.48 / 4.00 — *Los Angeles, CA*

SRM Institute Of Science And Technology — June 2018 – May 2022
Bachelor of Technology in Computer Science Engineering – GPA : 9.34 / 10.00 — *Chennai, India*

Experience

Software Engineer — August 2025 – Present
Easley-Dunn Productions, Inc. — *Sunnyvale, CA*

- Architecting MVC subsystem and optimized data layer, reducing storage by 40% (<500KB) and halving load times
- Implemented object pooling to boost mobile rendering performance by 33% on resource-constrained devices
- Engineering automated testing framework (EditMode/PlayMode) covering 200 tests, cutting regression bugs by 60%

Machine Learning Engineer — August 2023 – December 2023
USC Information Sciences Institute — *Los Angeles, CA*

- Fine-tuned RoBERTa on 349k+ questions using targeted oversampling, improving minority-class recall by 20% and boosting overall F1-score from 0.75 to 0.82
- Engineered scalable Python/PyTorch pipelines to process 300k+ records, reducing data preparation time by 40%

Software Engineer — August 2021 – July 2023
Highradius — *Hyderabad, India*

- Architected a centralized Data Catalog (Java/Spring Boot) to normalize 30M+ records monthly from 30+ banks, CRM and legacy ERPs, building ETL pipelines that established a unified source of truth
- Engineered a Server-Driven UI architecture (React, JSON Schema) to render dynamic dashboard and forms at runtime, abstracting tenant logic to reduce code duplication by 40% and eliminate custom client builds
- Scaled anomaly detection microservices to achieve 97% accuracy, driving 5 enterprise renewals and 20 new client
- Spearheaded a team to build a Selenium/Jenkins CI/CD automation framework, delivering 750+ tests 3 months early to reduce manual QA by 60% and block 35% of pre-merge bugs

Projects

Transcribe AI | *(Link)* | *React, TypeScript, Node.js, WebSockets, gRPC, Google Cloud Speech-to-Text*

- Architected a real-time Chrome Extension with a Node.js backend, utilizing a WebSocket-to-gRPC pipeline to reduce transcription latency to <300ms via Google Cloud API
- Developed a secure React/TypeScript side-panel UI with a backend proxy, achieving 100% credential isolation.

ClimaView | *(Link)* | *React, Redux, Node.js, Express, MongoDB, Google Cloud Platform, Tomorrow.io*

- Achieved sub-second latency by architecting MERN app using Tomorrow.io and Google Places APIs for live forecasts
- Engineered a Redux-driven React frontend with Highcharts to visualize complex datasets, supported by optimized Node.js/MongoDB endpoints that minimize redundant fetches.

Deep Learning Regularization Optimization for Image Classification | *(Link)* | *TensorFlow, Keras, Scikit-learn*

- Designed and trained 5-layer CNN from scratch using Keras & TensorFlow to classify images on the ImageNet dataset.
- Achieved a 15.4% reduction in validation loss (2.35 to 1.98) by developing a novel Ensemble Regularization strategy (L2/Dropout) and a robust Data Augmentation pipeline that bridged the generalization gap.

Enhanced Logical Reasoning Engine | *(Link)* | *PyTorch, Hugging Face Transformers, Scallopy (Neurosymbolic AI)*

- Engineered a neuro-symbolic DSR-ALBERT model to solve complex Kinship Classification, integrating symbolic reasoning to outperform baselines by 93% (19.1% to 37.0% accuracy) where commercial LLMs struggle
- Implemented a differentiable logic layer using Python, PyTorch, and Scallopy to autonomously learn integrity constraints, managing the end-to-end ML lifecycle from data preprocessing to evaluation

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash/Shell, Java, C++, HTML, C#, Swift
Frameworks & Libraries: Spring Boot, Node.js, React.js, Redux, PyTorch, TensorFlow, Keras, Scikit-learn, gRPC, WebSocket, GraphQL, Hibernate
Infrastructure & DevOps: AWS (EC2, Lambda, S3), GCP (App Engine), Docker, Kubernetes, Jenkins, Nginx, Git
Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB
Concepts: Distributed Systems, Microservices Architecture, System Design, CI/CD, RESTful APIs, Agile/Scrum